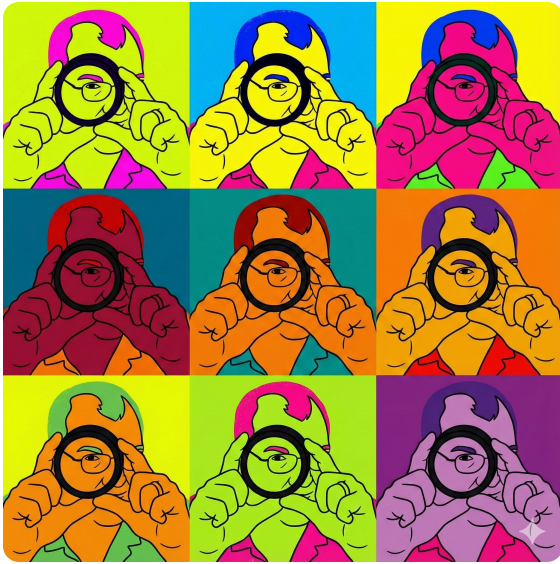




Montage Maker

An ImageMagick Companion

by Ed Johnson

User Guide

Montage Maker takes a folder full of images and turns it into a polished grid, contact sheet, social post, documentation image, or artful montage. It is useful for AI image batches, design variations, project photos, product mockups, reference sheets, and any visual collection that deserves to look organized instead of scattered.

You choose the grid size, tile dimensions, spacing, output format, text options, and visual effects. Montage Maker handles the assembly work using ImageMagick under the hood, then saves the result as one or more finished image files.

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What Is Montage Maker?

Montage Maker is a desktop tool for turning a folder of images into a clean, shareable montage. Point it at an image folder, choose a preset or customize the layout, and generate one or more finished output files.

It's great for:

- Creating contact sheets of AI image sets, design studies, or photo batches

- Building social-ready posts for Instagram, Pinterest, X/Bluesky, YouTube thumbnails, and link previews
- Comparing variations side by side so you can quickly spot your favorites
- Generating documentation images for 3D printing, laser projects, CAD workflows, tutorials, and product examples
- Making quick overview sheets of any image collection

[SCREENSHOT: The Montage Maker main window — use 2026-06-17_16-07-41.png]

Quick Start (TL;DR)

If you want to just jump in, here's the five-second version:

1. After installation (see instructions below), launch the app:
 - **Windows standalone:** double-click `Montage_Maker.exe`
 - **macOS standalone:** double-click `Montage_Maker`
 - **Python/source version:** run `venv_win\Scripts\python app.py` on Windows or `venv_mac/bin/python app.py` on macOS/Linux
2. In the **Left column**, click **Browse...** next to **Image folder** and pick your image folder.
3. Set your **Output folder** the same way, or leave it blank and Montage Maker will create/use a `Montages` folder inside your image folder.
4. In the **Middle column**, pick a **Preset** from the dropdown. Try **Square 3×3** to start.
5. Expand **Input Image Files** if you want to choose exactly which images are included.
6. Hit **Generate Montage**.
7. Your montage file(s) appear in the output folder, and a preview with an “X montage(s) created” count appears below the button.

That's it. Everything else in this guide is about fine-tuning.

Installation Options

There are two practical ways to use Montage Maker:

1. **Standalone app — easiest for most users.** Use the provided `Montage_Maker.exe` on Windows or `Montage_Maker` on macOS. You do not need to install Python for this option.

2. **Python/source version — best for tinkers, contributors, and anyone who wants to modify the code.** Use `setup.py` once, then run `app.py` from the virtual environment.

Either way, Montage Maker still needs **ImageMagick** installed and available on your system PATH, because ImageMagick does the actual montage-building work.

Recommended: Use the Standalone App

This is the easiest path for most users.

Windows

1. Install ImageMagick using the instructions below.
2. Put `Montage_Maker.exe` in a folder where you have permission to run apps.
3. Double-click `Montage_Maker.exe`.
4. If Windows asks for permission to run the app, approve it if you trust the source.

macOS

1. Install ImageMagick using the instructions below.
2. Put `Montage_Maker` somewhere convenient, such as your Applications folder or a project folder.
3. Double-click `Montage_Maker`.
4. If macOS blocks the app because it came from outside the App Store, use the normal macOS security workflow to allow it.

Note: The standalone app includes the Python application code, but it does **not** replace ImageMagick. ImageMagick still has to be installed separately and available on PATH.

Required: Install ImageMagick

Montage Maker uses **ImageMagick** to do the heavy lifting. Install it before running the app.

Windows 10 / 11

1. Download the installer from imagemagick.org — choose the 64-bit DLL build.
2. Run the installer.
3. Check “**Add application directory to your system path.**” This step matters.
4. Open a new terminal and type:

```
montage -version
```

You should see a version number.

macOS

```
brew install imagemagick
montage -version
```

Linux (Ubuntu / Debian)

```
sudo apt update && sudo apt install imagemagick
montage -version
```

Optional: Run from Python Source

Use this route if you want to run the source code directly, customize the app, or contribute changes.

Run the bootstrap script once with your system Python. It creates a virtual environment and installs Python dependencies.

```
# Windows
python setup.py

# macOS / Linux
python3 setup.py
```

After setup completes, launch the app with:

```
# Windows
venv_win\Scripts\python app.py

# macOS / Linux
venv_mac/bin/python app.py
```

Optional: Build Your Own Executable

Most users do not need this. If you are distributing Montage Maker or want to package your own standalone build, use the build instructions in `readme.md`.

The short version is:

```
# Windows
venv_win\Scripts\python build.py

# macOS
venv_mac/bin/python build.py
```

The build process creates `dist/Montage_Maker.exe` on Windows or `dist/Montage_Maker` on macOS. The executable still requires ImageMagick on PATH.

Running the App

Standalone app

```
Windows: double-click Montage_Maker.exe
macOS:   double-click Montage_Maker
```

A native desktop window opens — no browser required.

Python/source version

```
# Windows
venv_win\Scripts\python app.py

# macOS / Linux
venv_mac/bin/python app.py
```

CLI (for command-line fans)

The original command-line engine is still fully functional if you prefer that workflow. See [Using the CLI](#) at the end of this guide.

The Interface at a Glance

The app uses a three-column layout:

Column	What's in it
Left (~40%)	Image folder, Output folder, Input Image Files selection panel, Generate Montage button, and output previews
Middle (~25%)	Preset selector, Grid, Tile size & spacing, Background, Output format
Right (~35%)	Output filename prefix, Text (title, font, colors, sizes, labels), Effects

[SCREENSHOT: Annotated version of 2026-06-17_16-07-41.png with callouts for each column]

The **header** has two controls in the top-right corner:

Control	What it does
Theme dropdown	Switch between four themes: Light and Warm (light modes) or Dark and Midnight (dark modes). Your choice is saved automatically and restored on next launch.
Port: 8000 button	Change the server port if another app is using 8000. Takes effect after restarting the app.

Settings Walkthrough

Grid

Controls how many tiles appear on each page.

Field	What it does
Columns	Number of tiles per row
Rows	Number of rows per page

If your image folder has more images than fit on one page, Montage Maker automatically creates additional pages — `prefix_01.png` , `prefix_02.png` , and so on.

Example: A 3×3 grid with 12 images produces two pages — the first with 9 images, the second with 3.

[EXAMPLE OUTPUT: 3×3 grid of puppy images — 9 per page]

Tile Size & Spacing

Controls how big each individual tile is and how much breathing room sits between them.

Field	Format	What it does
Width / Height	pixels	The dimensions of each tile
H spacing / V spacing	pixels	The gap between tiles (horizontal and vertical)

Tip: Larger spacing gives your montage an airy, editorial feel. Setting both H and V spacing to 0 (as shown in the Square 3×3 preset) makes tiles sit flush against each other — clean and compact. The **Concatenate** mode under Effects does the same thing automatically.

[EXAMPLE OUTPUT: Same 3×3 grid — left with 10px spacing, right with 0px spacing / Concatenate]

Background

Field	What it does
Color	The fill color that shows between and behind tiles. Accepts hex values (#ffffff) or named colors (black , white , gray , etc.)
Transparent	Removes the background entirely. Output format is automatically forced to PNG when this is on, since JPG doesn't support transparency.

[EXAMPLE OUTPUT: White background vs. black background vs. transparent (checkerboard)]

Output

Field	What it does
Crop	Center-crops each source image to the specified size before tiling (e.g. 1080x1080). Great for making sure all tiles are the same proportions.
Format	The file format for output files: png , jpg , bmp , or tiff .

Tip: Use **Crop** when your source images have mixed aspect ratios and you want a clean, uniform grid. Without it, ImageMagick will try to fit each image into the tile space as-is, which can result in letterboxing.

Output Filename Prefix

The base name for your generated files. Pages are numbered automatically.

Example: A prefix of `puppies` with a 3x3 grid across two pages produces `puppies_01.png` and `puppies_02.png` .

Text (Titles & Labels)

Field	What it does
Title	A text banner printed above the entire montage
Font name	Font for all text — type to filter from a searchable list of installed fonts
Text color	Fill color for both the title and filename labels. Default is black — change this when your background is dark.
Title size	Point size for the title banner (independent of the label size)
Label size	Point size for filename labels
Show filename labels	Adds the source filename below each tile — great for contact sheets and documentation

Tip: Title size and label size are fully independent — you can have a large, bold title at 48pt while keeping compact 10pt labels below each image.

Tip: The **Contact Sheet** preset has labels on by default — it's a handy reference for what labeled output looks like.

[EXAMPLE OUTPUT: Contact sheet with filename labels under each tile]

Effects

This is where things get fun.

Control	What it does
Mode	Frame — adds a decorative 3D border around tiles. Concatenate — zero spacing, tiles touch edge-to-edge.
Quality (JPG)	JPEG compression quality, 0–100. Default is 85. Has no effect on PNG output.
Drop shadow	Adds a subtle shadow behind each tile. Automatically disabled while Polaroid is active — Polaroid already bakes in its own shadow.
Polaroid	Applies a Polaroid-style effect. Set a fixed angle, or enable Random (±15°) for a casually tossed look.
Frame	A beveled ornamental border around each tile. Controls: Width, Height, Outer bevel, Inner bevel, and Matte color for the frame color. Pairs naturally with Mode = Frame.
Border	A flat solid border around each tile. Controls: Width, Height, and Border color .

[EXAMPLE OUTPUT: Drop shadow effect on a 2×2 grid]

[EXAMPLE OUTPUT: Polaroid effect with Random angle enabled]

[EXAMPLE OUTPUT: Frame effect with gray matte color]

Tip: Drop shadow and Polaroid look especially nice together for a "scattered photos" aesthetic.

Selecting Which Images to Include

Just above the **Generate Montage** button is a collapsible **Input Image Files** panel. Expand it to see every image in your input folder, each with a small thumbnail preview and a checkbox.

- Check the images you want included — unchecked images are skipped entirely.
- Use **All** / **None** to select or deselect everything at once.
- The count in the header (e.g. "7 of 12 selected") stays visible even when the panel is collapsed, so you always know your current selection at a glance.
- The list refreshes automatically when you change the input folder.

Tip: This is much faster than moving files in and out of the folder when you want to experiment with different subsets of your images.

Presets — Your Best Friends

Presets are saved collections of all your settings. Instead of dialing in 15 different fields, you just pick a preset from the dropdown and you're ready to go. Montage Maker ships with eleven presets built in:

Preset	Grid	Description
Contact Sheet	4×4	300px tiles with filename labels, cropped square — classic proof sheet
Filmstrip	4×1	480px tiles, Concatenate mode, black background, proportional black borders — looks like a real film strip
Gallery Wall	2×2	500px tiles, beveled frame, drop shadow, warm linen background — framed prints on a wall
Instagram Post	2×2	Cropped to 1080×1080, JPG — ready to upload
Instagram Story	1×1	Cropped to 1080×1920, JPG — full-screen story / TikTok format
Link Preview	2×2	Cropped to 1200×630, JPG — the universal OG image format for Discord, Slack, and social link shares
Pinterest Pin	2×3	Cropped to 1000×1500, JPG — tall Pinterest format

Preset	Grid	Description
Polaroid	3×2	450px tiles, Polaroid effect with random angles, dark background and drop shadow
Square 3×3	3×3	720px tiles, PNG — a versatile general-purpose layout
X Post	2×2	Cropped to 1200×675, JPG — X / Bluesky ready
YouTube Thumb	2×2	Cropped to 1280×720 — YouTube thumbnail dimensions

[EXAMPLE OUTPUT: Instagram Post 2×2 — cropped square tiles]

[EXAMPLE OUTPUT: Contact Sheet 4×4 with filename labels]

[EXAMPLE OUTPUT: Pinterest Pin 2×3 — tall format]

[EXAMPLE OUTPUT: Square 3×3 — the puppy sample image you already have!]

Tip: The social media presets (Instagram, Pinterest, X Post, Link Preview, YouTube Thumb) all have crop values set to their platform's ideal dimensions. This means your output is genuinely ready to upload — no resizing needed.

The Preset Manager

Click **Manage...** next to the Preset dropdown to open the Preset Manager.

From here you can:

- **Load** an existing preset into the form to review or edit it
- **Save** under any name you like (spaces and proper case are fine; the only characters that aren't allowed are `]` and newlines)
- **Delete** a preset you no longer need
- **Clear** the form to start fresh from scratch

Presets are stored in `config.ini` in the same folder as the script, and are sorted alphabetically in the dropdown.

[SCREENSHOT: Preset Manager dialog]

Tip: After dialing in a custom setup you like — say, your preferred tile size, a specific background color, and drop shadow on — save it as a preset with a memorable name. Future-you will be grateful.

File Conflict Dialog

If files with your chosen prefix already exist in the output folder, a dialog pops up before anything is written. You have three choices:

Option	What happens
Cancel	Nothing is written. You're back where you started.
Auto-increment	Appends <code>_v2</code> , <code>_v3</code> , etc. to create a new set alongside the existing files.
Overwrite	Replaces the most recent version only (e.g. if <code>_v3</code> files exist, only those are replaced — earlier versions are left alone).

The dialog tells you exactly how many files are affected and which prefix Overwrite will target, so there are no surprises.

Tip: Auto-increment is the safe default if you're experimenting. Overwrite is handy when you're iterating on a final result and don't want to accumulate numbered versions.

Using the CLI

Prefer the command line? The original CLI engine is still fully functional and lives in `montage_maker.py` .

```
python montage_maker.py [GRID] [OPTIONS]
```

Arguments

Argument	Default	Description
GRID	2x2	Grid layout, e.g. <code>3x4</code> . Optional if <code>--preset</code> is used.

Argument	Default	Description
--preset NAME	—	Load all settings from a named <code>config.ini</code> section
--ext EXT	png	Output file format
--size WxH+HB+VB	500x500+10+10	Tile size + horizontal/vertical spacing
--label	off	Turn filename labels on
--fontsize N	12	Label font size in points
--prefix NAME	montage	Output filename prefix
--crop WxH	—	Center-crop each image before tiling

Examples

```
# Simple 2x2 grid with default settings
python montage_maker.py 2x2
```

```
# Use a saved preset
python montage_maker.py --preset "Instagram Post"
```

```
# Custom 2x2 with specific tile size, labels on, and larger font
python montage_maker.py 2x2 --size "500x500+5+5" --label --fontsize 24
```

Note: The CLI scans the **current working directory** for images, so `cd` into your image folder (or the folder containing them) before running. Output goes to the current directory as well. A `process.log` is written alongside the output files.

Troubleshooting

montage: command not found

ImageMagick isn't on your PATH. Reinstall it with "Add to PATH" checked during setup, then open a brand new terminal window.

The app window fails to open (Windows)

The native window needs the Microsoft Edge WebView2 runtime. It comes pre-installed on Windows 10 and 11, but if it's missing, grab it from developer.microsoft.com/microsoft-edge/webview2.

"Connection lost" toast appears during generation

This shouldn't happen in the current version (generation runs in a background thread). If it does appear, it means ImageMagick is taking an unusually long time with a large batch. Give it a moment — it should finish.

Port conflict on startup

Click **Port: 8000** in the app header, enter a different port number that's not in use, save, and restart the app.



Montage Maker is built with [NiceGUI](#) and [ImageMagick](#).

♥ Support the Maker (and Lucy!)

I develop these tools to improve my own workflows and love sharing them with the community. If you find Montage Maker useful and want to say thanks, feel free to [buy Lucy a dog treat on Ko-fi!](#)

Happy Making!

— EdJ